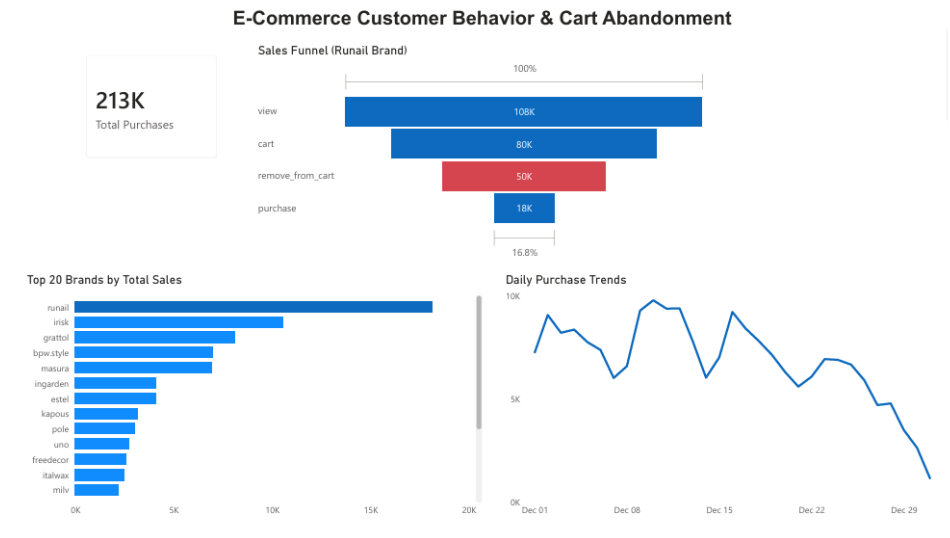


Project: E-Commerce Customer Behavior & Cart Abandonment Analysis



Executive Summary & Insights

•**The Challenge:** Diagnosed high cart abandonment in a cosmetics e-commerce dataset containing over 3 million interaction records. Utilized Google BigQuery and Power BI to analyze the sales funnel and identify critical friction points.

•**Key Insight 1 (The "Price-Shipping" Friction):** Contrary to the initial hypothesis that high prices deter purchases, drill-down analysis revealed that the most abandoned products were extremely low-priced (e.g., \$0.40). This indicates customers are abandoning carts when minor purchases are overshadowed by disproportionate checkout/shipping fees.

•**Key Insight 2 (The 80/20 Rule):** A single brand ("runail") drives the majority of purchases but suffers from a massive abandonment rate (80.4K adds to cart vs. 49.5K removals).

•**Key Insight 3 (Temporal Drop):** Purchase behavior consistently declines in the final week of the month, aligning with consumer purchasing power cycles.

Business Recommendations:

• **Recommendation 1:** Implement dynamic shipping thresholds or introduce bundle offers for low-ticket items to reduce "checkout shock" and improve the conversion rate.

• **Recommendation 2:** Introduce targeted discounts or free shipping campaigns during the last week of the month to stabilize the end-of-month revenue drop-off.

• **Business Impact:** Resolving the checkout friction for the top-performing brand alone has the potential to recover thousands of lost transactions and significantly boost Overall Revenue

Technical Execution & Data Pipeline:

1. Methodology & Architecture: To bypass local hardware limitations when handling over 3 million rows of raw transactional data, I designed a cloud-based pipeline. I utilized **Google BigQuery (SQL)** to clean the data (filtering undefined variables), perform temporal trend analysis, and quantify the e-commerce funnel. The aggregated results were then connected to Power BI for executive visualization.

2. The SQL Engine (Advanced Drill-Down):

Below is a snippet of the strategic SQL queries used to isolate the root cause of cart abandonment for the top-performing brand.

```
/*
STEP 8: Funnel analysis for the top brand ('runail') to pinpoint drop-off points.
This query quantifies the exact volume of friction in the customer journey.
*/
SELECT
    event_name,
    brand,
    COUNT(event_name) AS interaction_count
FROM `data-analysis-projects-496119.cosmetics_ecommerce.events_raw`
WHERE brand = 'runail'
GROUP BY event_name, brand
ORDER BY interaction_count DESC;

/*
STEP 9: Identifying specific products causing cart abandonment.
This drill-down analysis isolated the exact product IDs causing the friction,
ultimately revealing the "Price-Shipping" mismatch.
*/
SELECT
    product_id,
    brand,
    event_name,
    COUNT(event_name) AS total_removals
FROM `data-analysis-projects-496119.cosmetics_ecommerce.events_raw`
WHERE brand = 'runail' AND event_name = 'remove_from_cart'
GROUP BY product_id, brand, event_name
ORDER BY total_removals DESC
LIMIT 20;
```